

Code-Layout, Readability and Reusability

Date	7 July 2026
Team ID	SWTID-2026-8075
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout	Yes	Uniform indentation maintained throughout the project.
2	Proper File Structure	Files and folders are logically organized	Yes	Organized into app.py, similarity.py and pdf_report.py modules.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Clear names like reference answers, pause percent, feedback.
4	Function / Method Names	Functions are descriptively named	Yes	Descriptive functions such as calculate similarity () and generated ()
5	Code Comments	Inline and block comments explain log	Yes	Section-wise comments improve readability.
6	Modular Design	Code is split into reusable functions/m	Yes	Separate modules for UI, similarity and PDF generation.
7	No Redundant Code	Duplicate or unused code is removed	Yes	No unnecessary duplicate code found.
8	Error Handling	Exceptions and errors are handled grab	Partil	Basic validation present; comprehensive try-except can be improved

Reusable Components / Modules:

S.No.	Component / Module Name	Language / Technology	where Reused	Reusability Level (High / Medium / Low)
1	Speech-to-Text Module	Python / Whisper	Audio Transcription	High

2	Semantic Similarity Module	Python	Concept Evaluation	High
3	Audio Analysis Module	Python / Libros	Fluency & Voice Analysis	High
4	Streamlet UI Components	Python / Streamlet	User Interface	Medium
5	PDF Report Generator	Python	Report Generation	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized modular project structure.
Readability	5	Clear naming conventions and section-wise organization.
Reusability	5	Modules can be reused in similar applications.
Documentation / Comments	4	Basic comments provided for major sections.
Overall Score	4.8	Good quality, maintainable and reusable code.